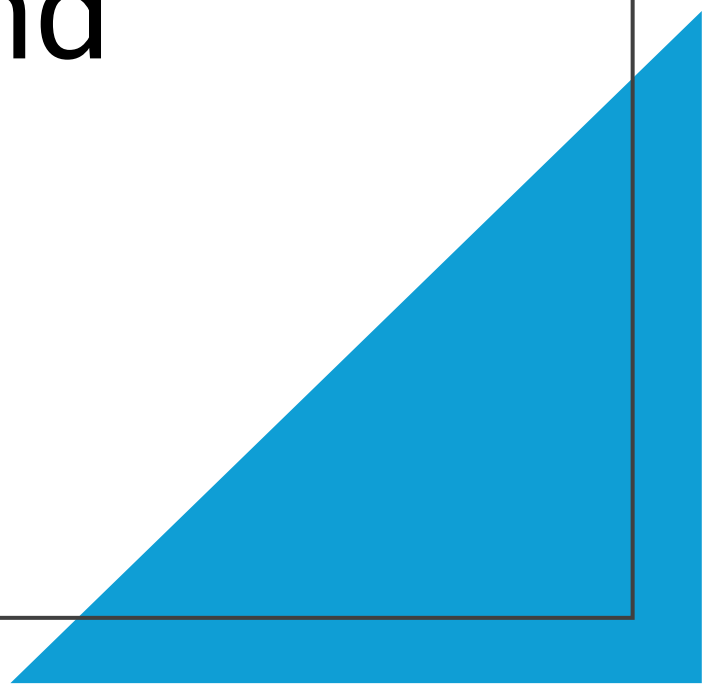


CDC NORS Outbreaks: Exploratory Analysis of Etiology, Transmission, Settings, and Health Outcomes

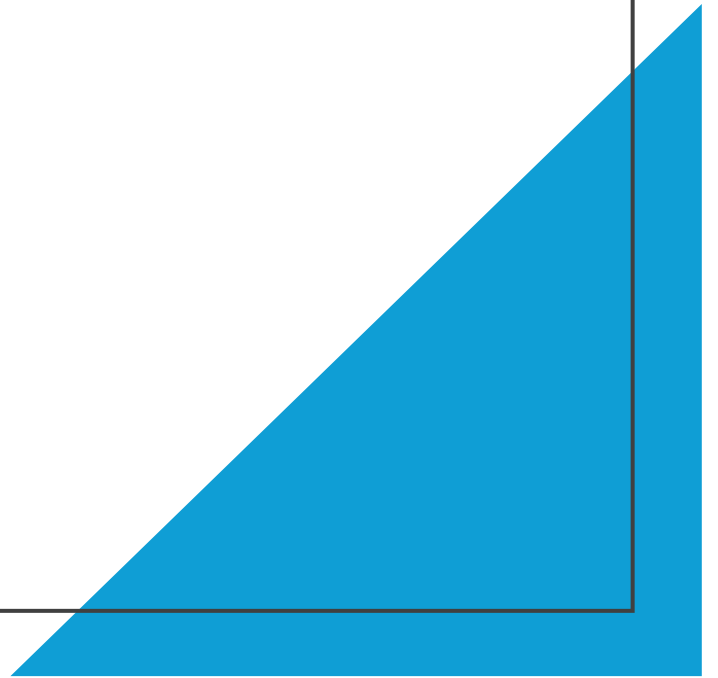
Alfadhul Almubarak

DSB-3

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Content

1. Problem Statement
2. Aim and Objectives
3. Keywords
4. Data Analysis
5. Limitations
6. Summary

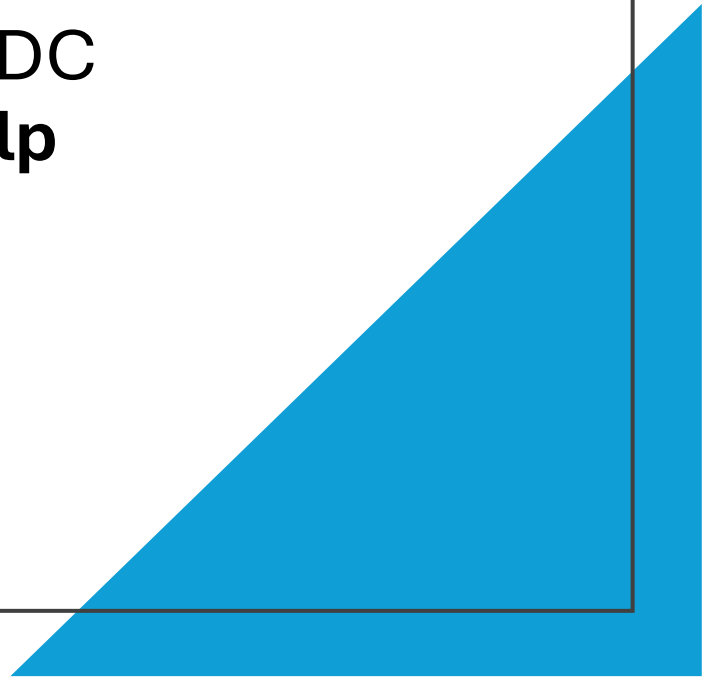


Problem Statement

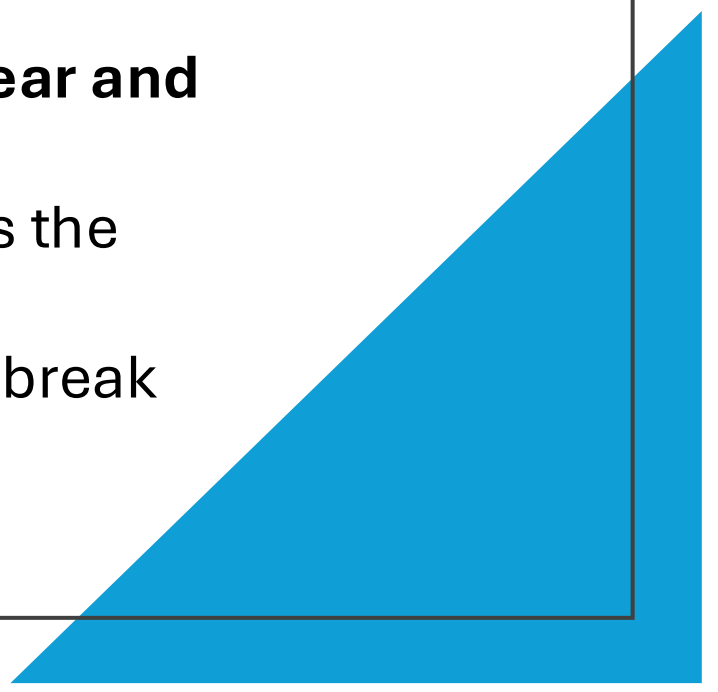
- The CDC National Outbreak Reporting System (NORS) contains reported foodborne, waterborne, and related disease outbreaks across the United States
- In the dataset covering **1971–2023**, for **24,889** outbreaks, **Norovirus** accounted for **52.75%** of reported outbreaks, compared with **15.50%** for **Salmonella**
- However, differences in outbreak frequency alone do not provide a complete understanding of the public-health burden
- Examining how **different etiologies vary** in transmission modes, outbreak settings, outbreak size, hospitalization, and mortality can help identify important patterns that **may inform public-health measures, infection-control practices, and food-safety interventions**

Aim

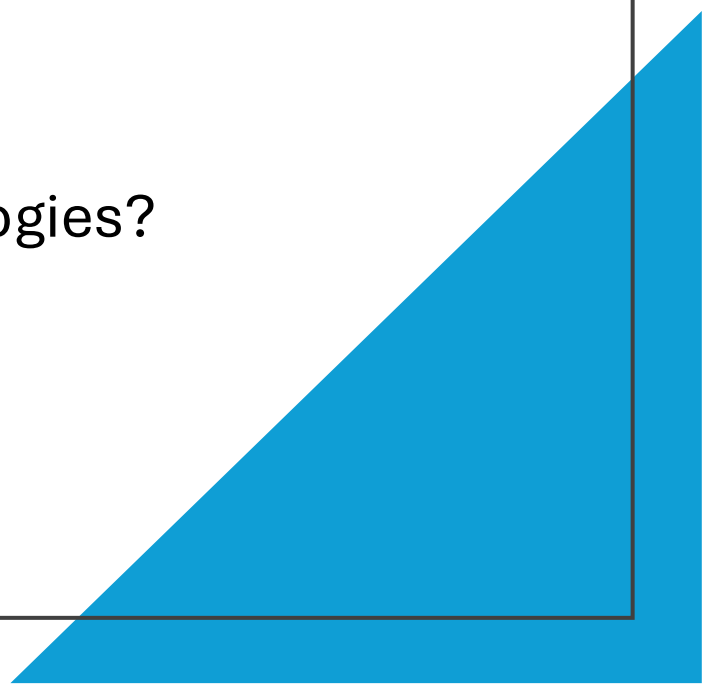
- To **explore patterns in reported disease outbreaks** across different etiologies using CDC NORS data, to **identify patterns that may help inform public-health measures, infection-control practices, and food-safety interventions**



Objectives

- What **proportion of reported outbreaks** was attributed to each major etiology?
 - How did the occurrence of these etiologies **vary by year and month**?
 - How did **primary modes of transmission** vary across the major outbreak etiologies?
 - How did **outbreak settings** vary across the major outbreak etiologies?
- 

Objectives

- What was the typical outbreak size for each major etiology, based on the **number of reported illnesses per outbreak**?
 - How did the proportion of outbreaks with at least one recorded **hospitalization** vary across the major etiologies?
 - How did the proportion of outbreaks with at least one recorded **death** vary across the major etiologies?
- 



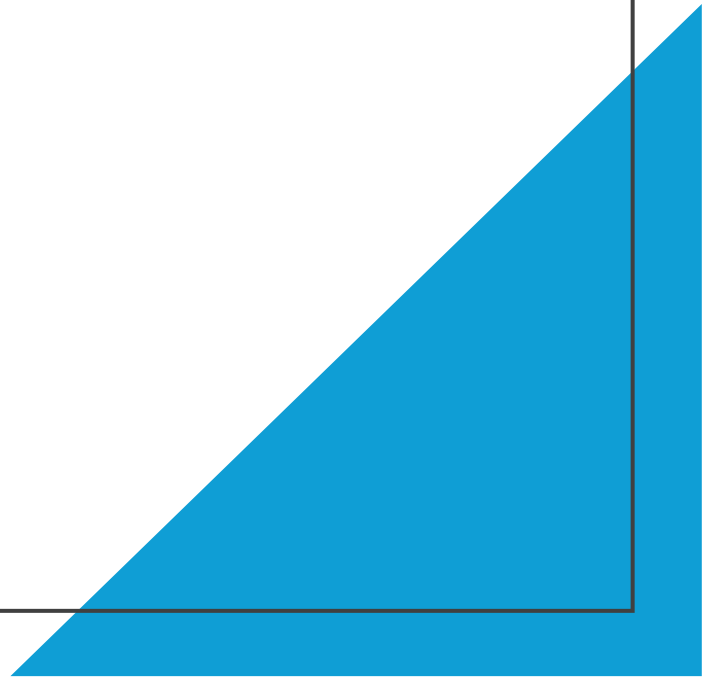
Definitions

Etiology: The specific cause or causative agent of a disease, such as a virus, bacterium, parasite, or toxin

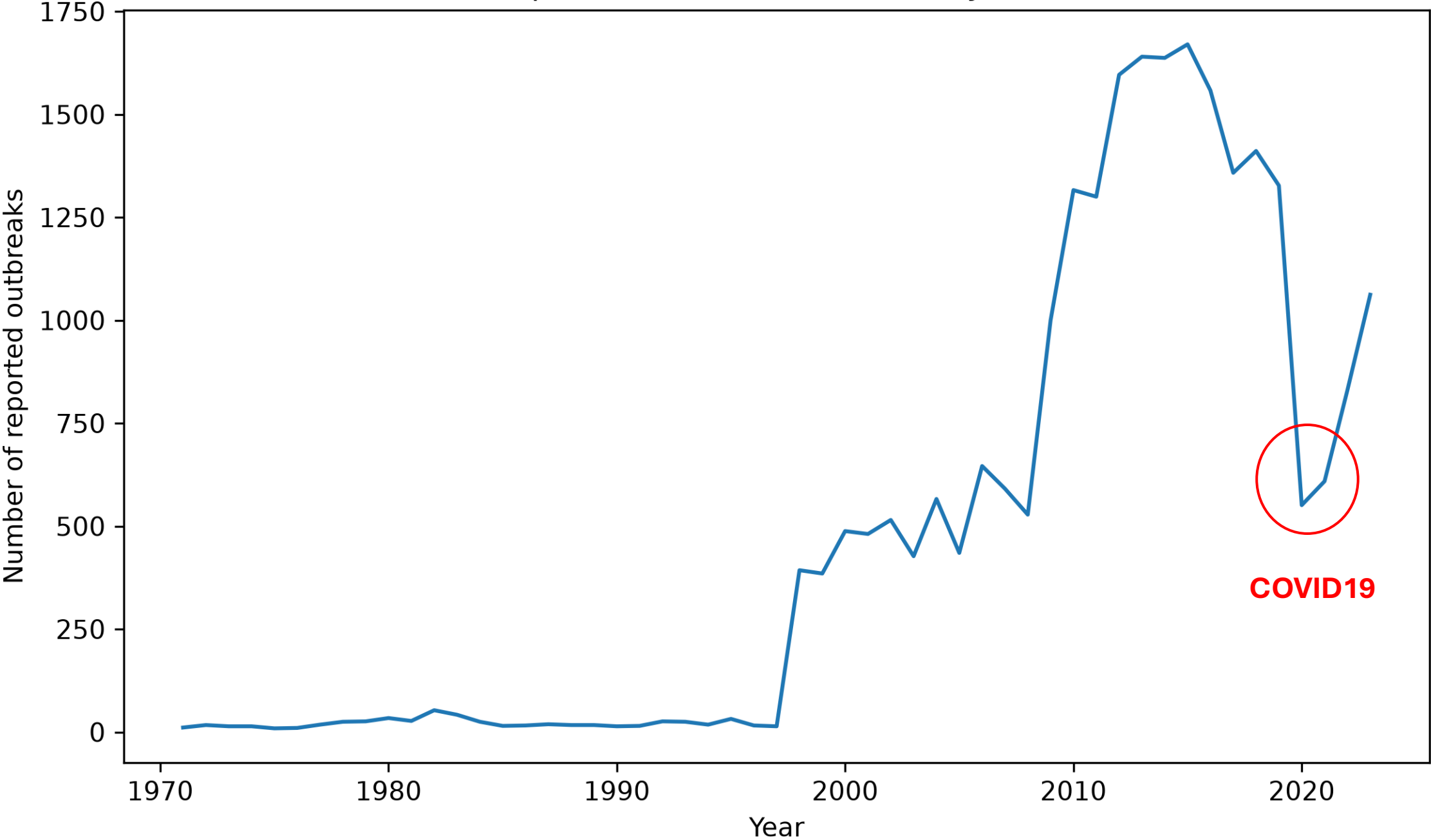
Outbreak: The occurrence of more cases of a disease than normally expected within a specific population, place, or period

Introducing the dataset

- **CDC NORS**
 - Reported outbreaks, United States, **1971–2023**
- **Analysis dataset**
 - Single etiology
 - Confirmed etiology only
 - Missing/multiple/suspected etiologies excluded
 - Top 10 etiologies + Other



Reported Disease Outbreaks by Year





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Letter to Editors

Reduction in the incidence of infectious diseases during the COVID-19 pandemic: A hypothesis

ARTICLE INFO

Keywords

COVID-19
SARS-COV-2
Infectious diseases
Reduction
Incidence

Public Health Measures During the COVID-19 Pandemic Reduce the Spread of Other Respiratory Infectious Diseases

Cheng-yi Hu¹, Yu-wen Tang¹, Qi-min Su¹, Yi Lei¹, Wen-shuai Cui¹, Yan-yan Zhang¹, Yan Zhou¹, Xin-yan Li¹, Zhong-fang Wang² and Zhu-xiang Zhao^{1*}

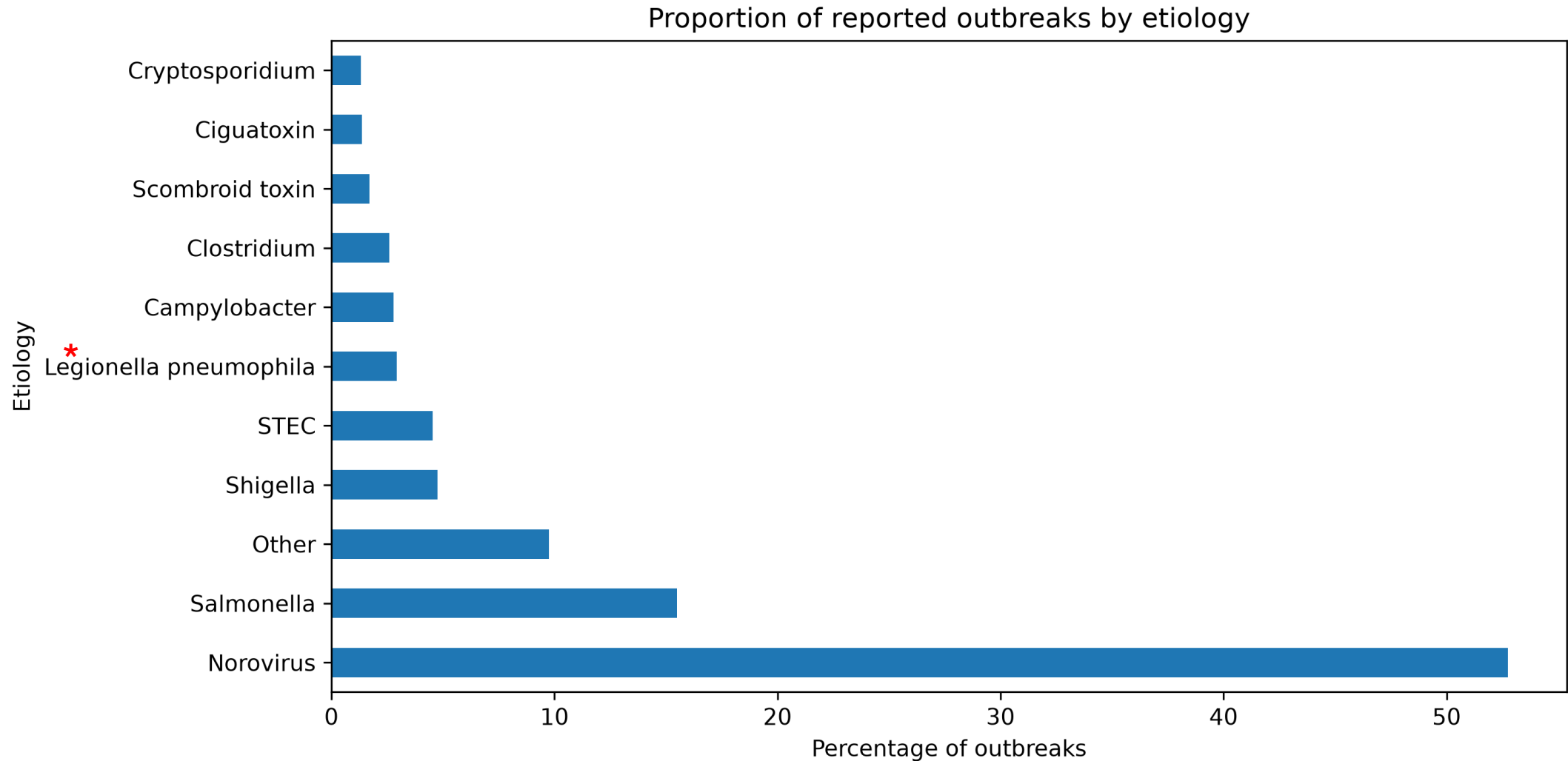
¹ Department of Infectious Diseases, Guangzhou First People's Hospital, School of Medicine, South China University of Technology, Guangzhou, China, ² State Key Laboratory of Respiratory Disease & National Clinical Research Center for Respiratory Disease, Guangzhou Institute of Respiratory Health, The First Affiliated Hospital of Guangzhou Medical University, Guangzhou Medical University, Guangzhou, China

Background: Public health measures (such as wearing masks, physical distancing, and isolation) have significantly reduced the spread of the coronavirus disease-2019 (COVID-19), but the impact of public health measures on other respiratory infectious diseases is unclear.

The decrease in the proportion of viral outbreaks and the proportion of outbreaks with a contamination contributing factor during 2020–2022 might be attributed to effects from the COVID-19 pandemic.

Nonpharmaceutical interventions (e.g., increased glove use, cleaning and disinfection, and closure of restaurant dining areas) implemented during the COVID-19 pandemic likely led to a reduction in norovirus, which is typically spread by infectious food workers

What proportion of reported outbreaks was attributed to each major etiology?

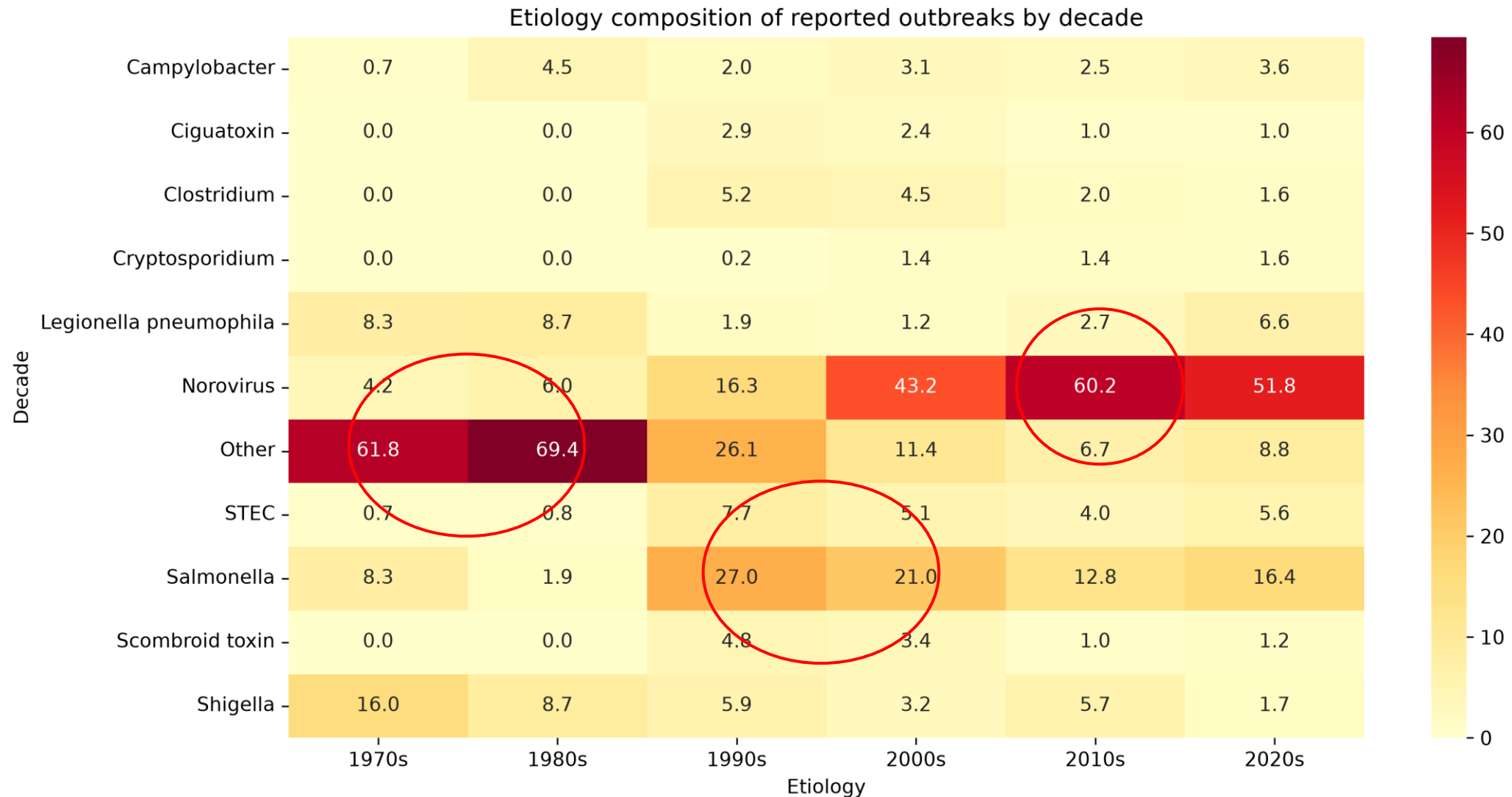


What **proportion of reported outbreaks** was attributed to each major etiology?

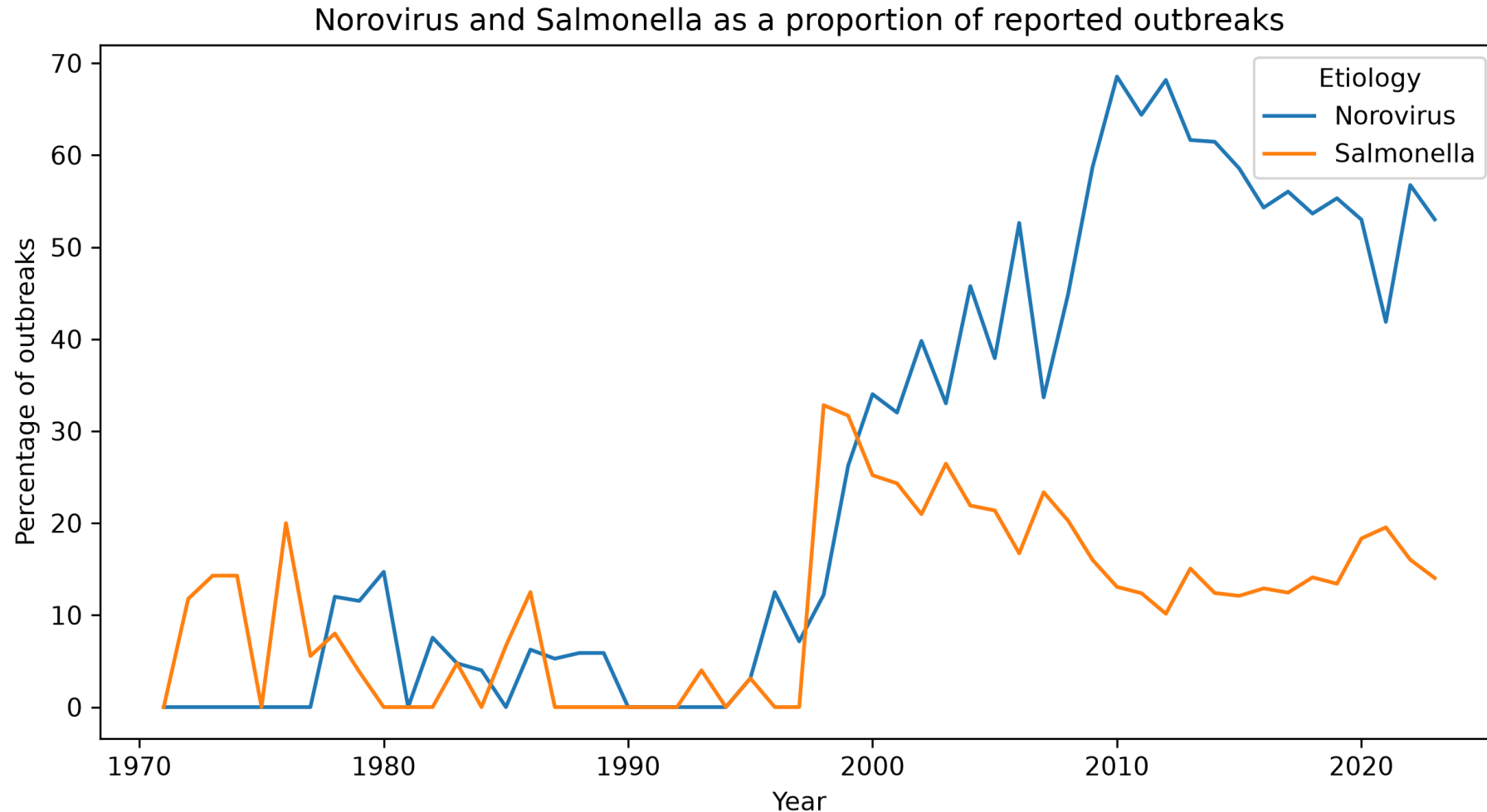
- **Viruses:** Norovirus
- **Bacteria:** Salmonella, Shigella, STEC (*Shiga toxin-producing E. coli*), Legionella pneumophila, Campylobacter, Clostridium
- **Parasites:** Cryptosporidium
- **Foodborne Toxins:** Scombroid toxin, Ciguatoxin
- **Other** — a combined category containing less frequent etiologies that weren't retained as individual categories



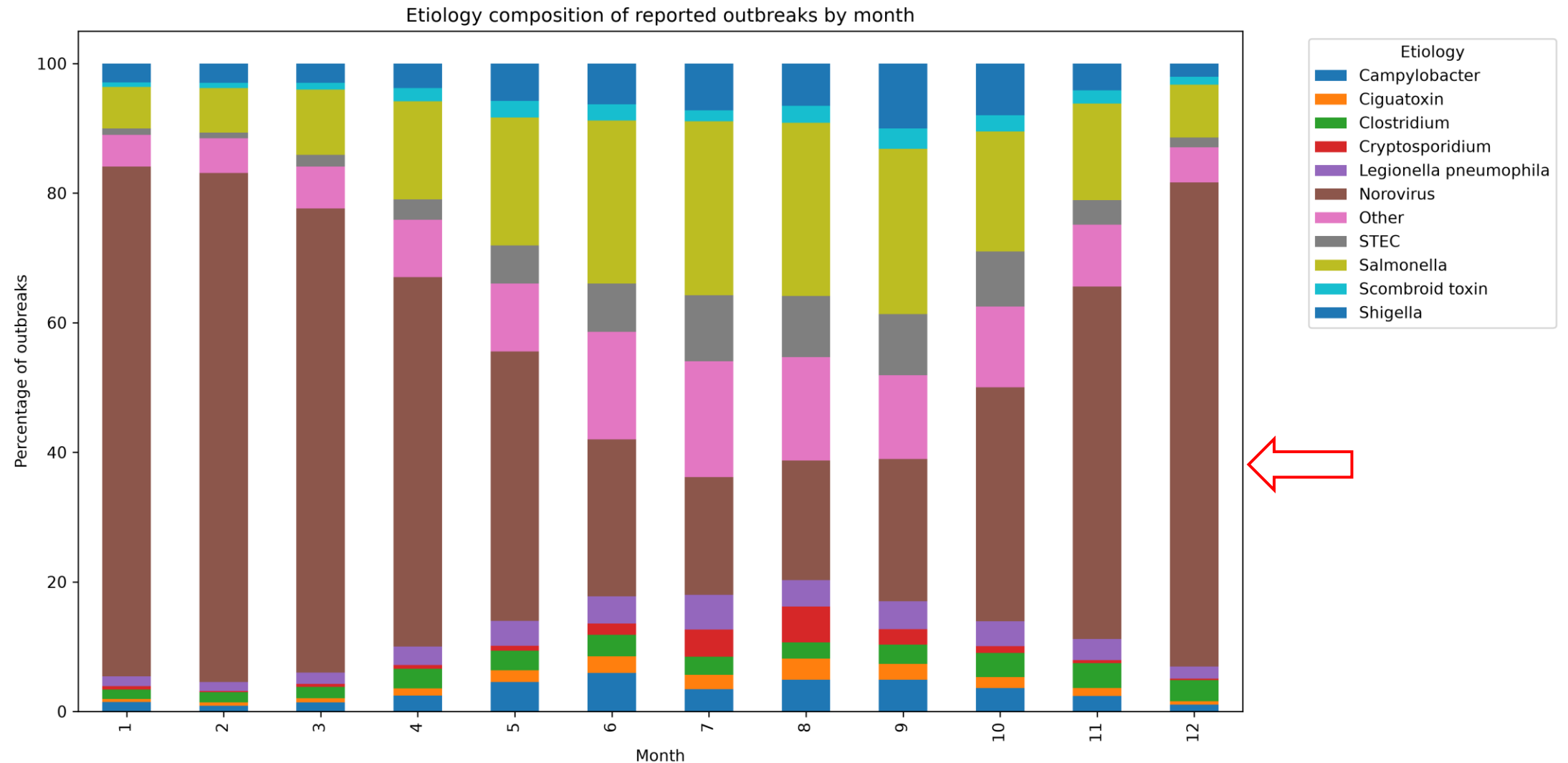
How did the occurrence of these etiologies vary by year and month?



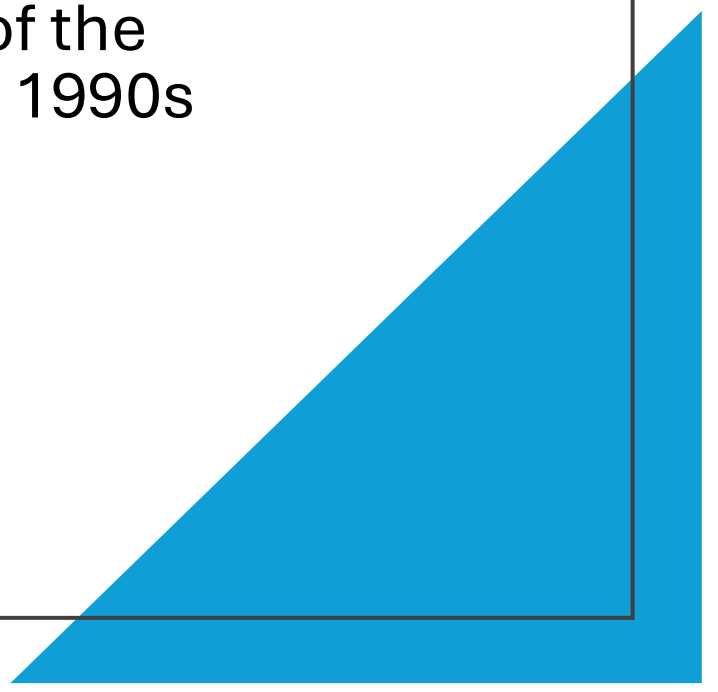
How did the occurrence of these etiologies **vary by year and month?**



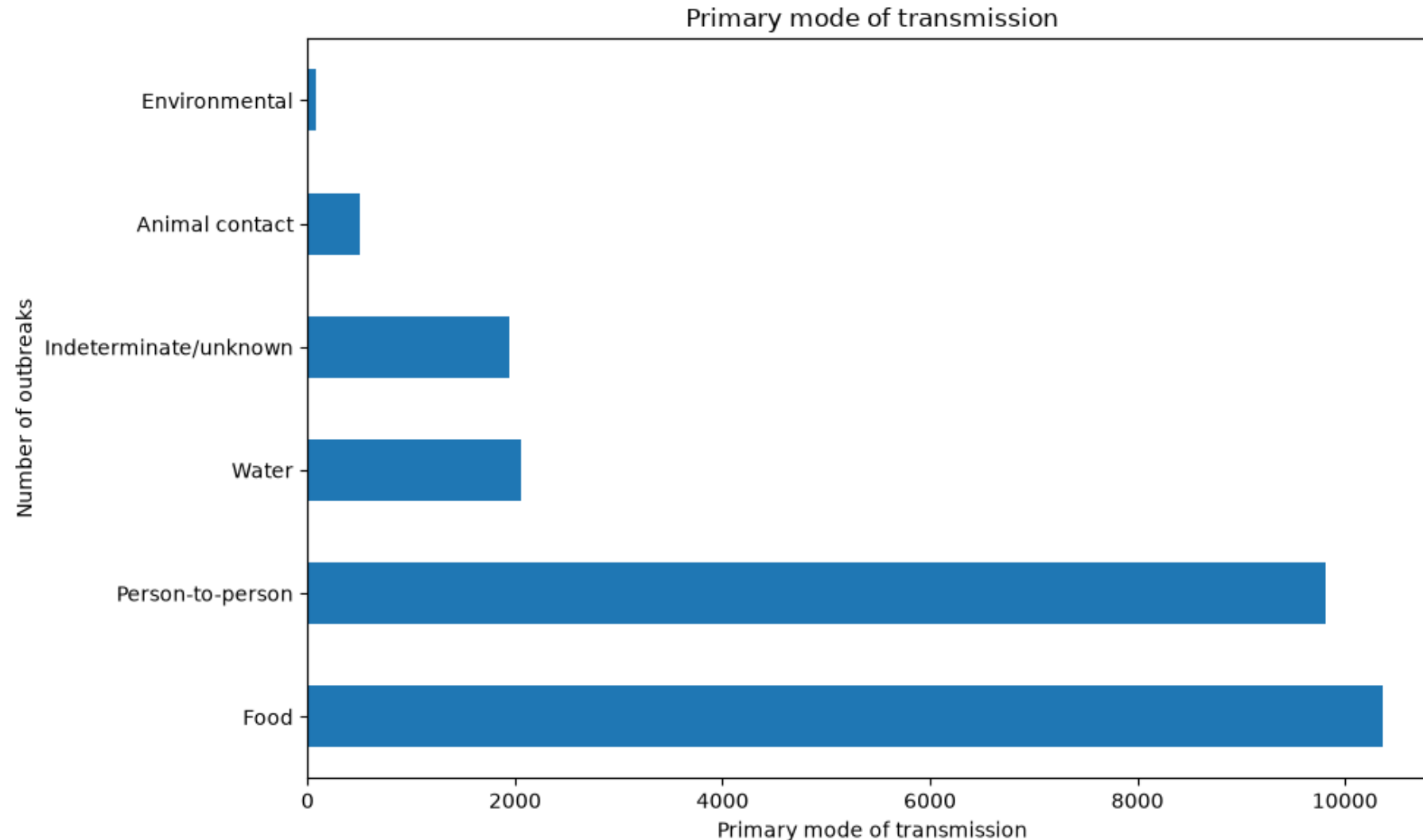
How did the occurrence of these etiologies vary by year and month?



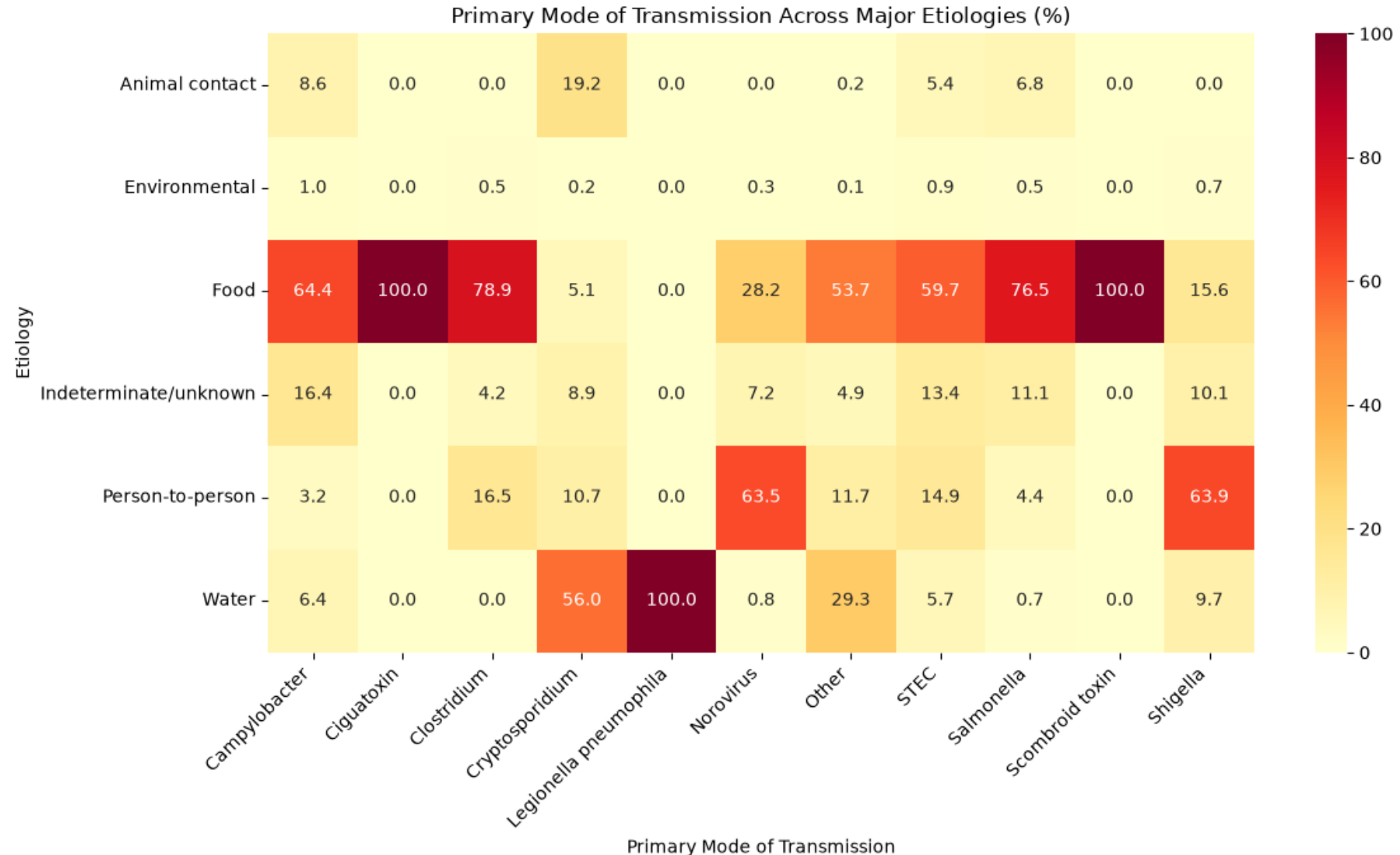
How did the occurrence of these etiologies **vary by year and month?**

- The number of reported outbreaks varied substantially over time
 - Reporting was relatively low during the earlier years of the dataset, followed by a marked increase from the late 1990s onward
 - Annual reporting reached its highest level in **2015**, with **1,670 outbreaks**, before declining in 2020 and subsequently increasing again through 2023
 - This demonstrates substantial temporal variation in reported outbreak occurrence
- 

How did **primary modes of transmission** vary across the major outbreak etiologies?



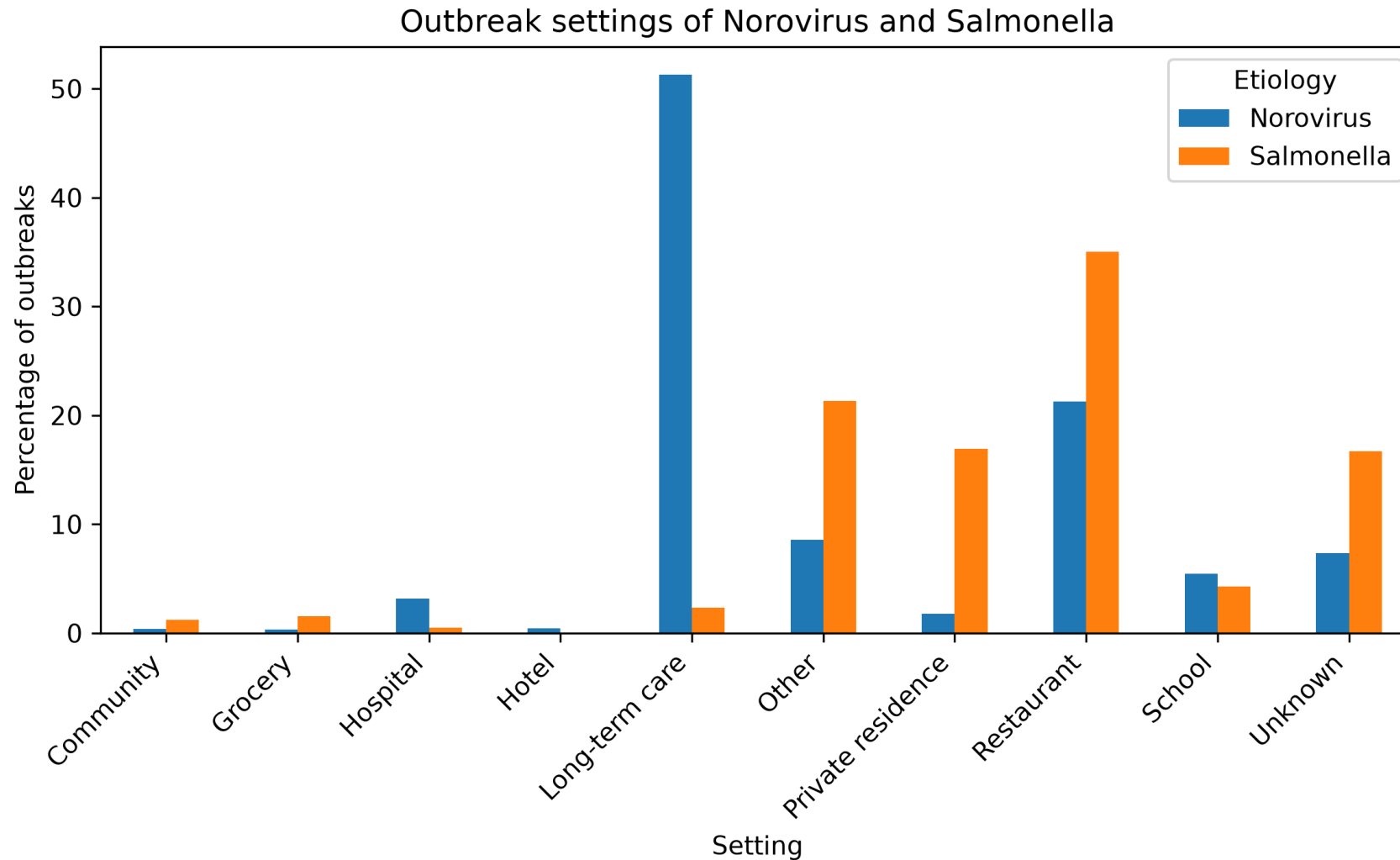
How did **primary modes of transmission** vary across the major outbreak etiologies?



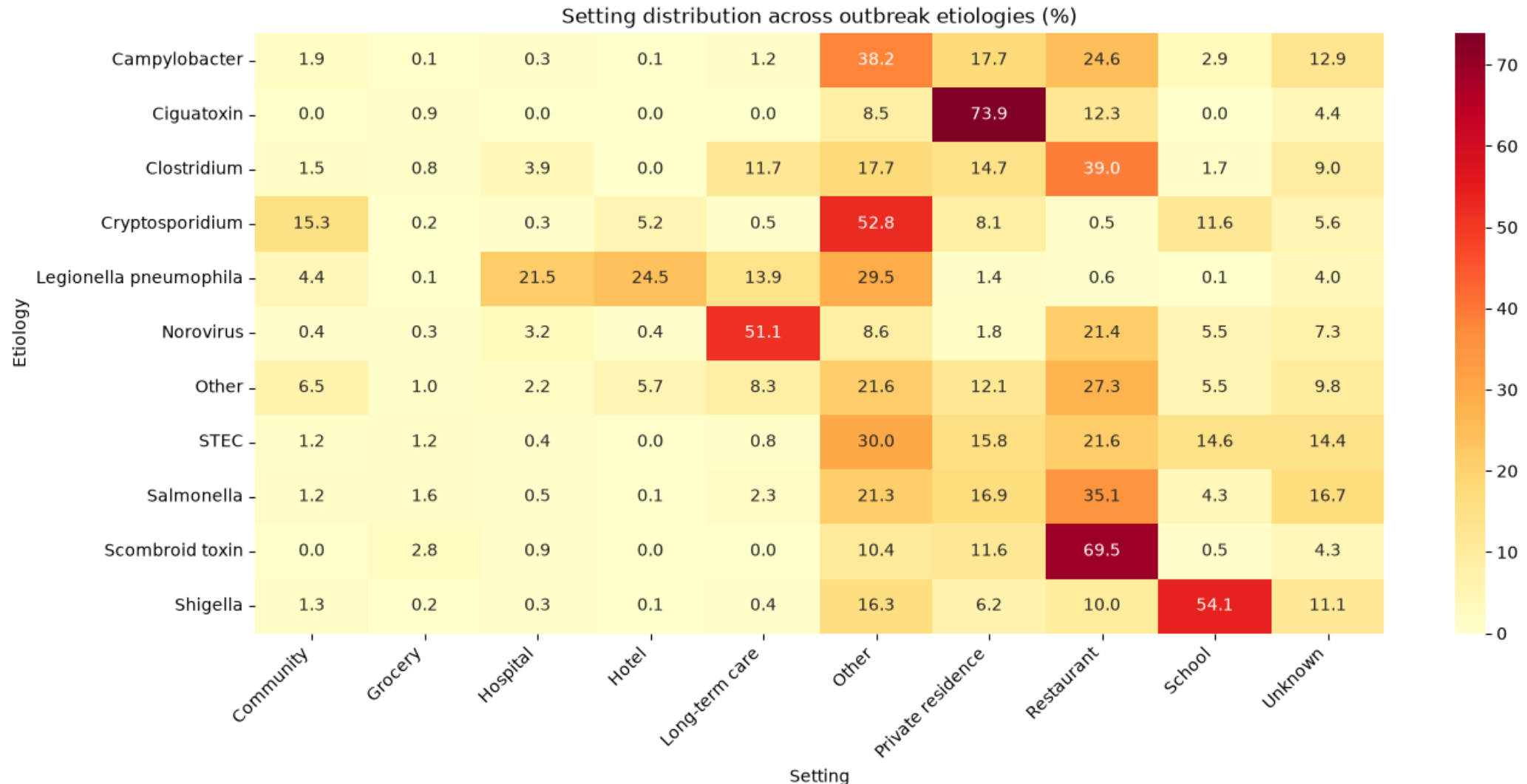
How did **primary modes of transmission** vary across the major outbreak etiologies?

- The pattern varied substantially by etiology
- For example, Norovirus outbreaks were strongly represented by person-to-person transmission, whereas Salmonella outbreaks had a substantial foodborne component
- Other etiologies showed distinct transmission profiles, associated with water or animal contact

How did **outbreak settings** vary across the major outbreak etiologies?



How did **outbreak settings** vary across the major outbreak etiologies?

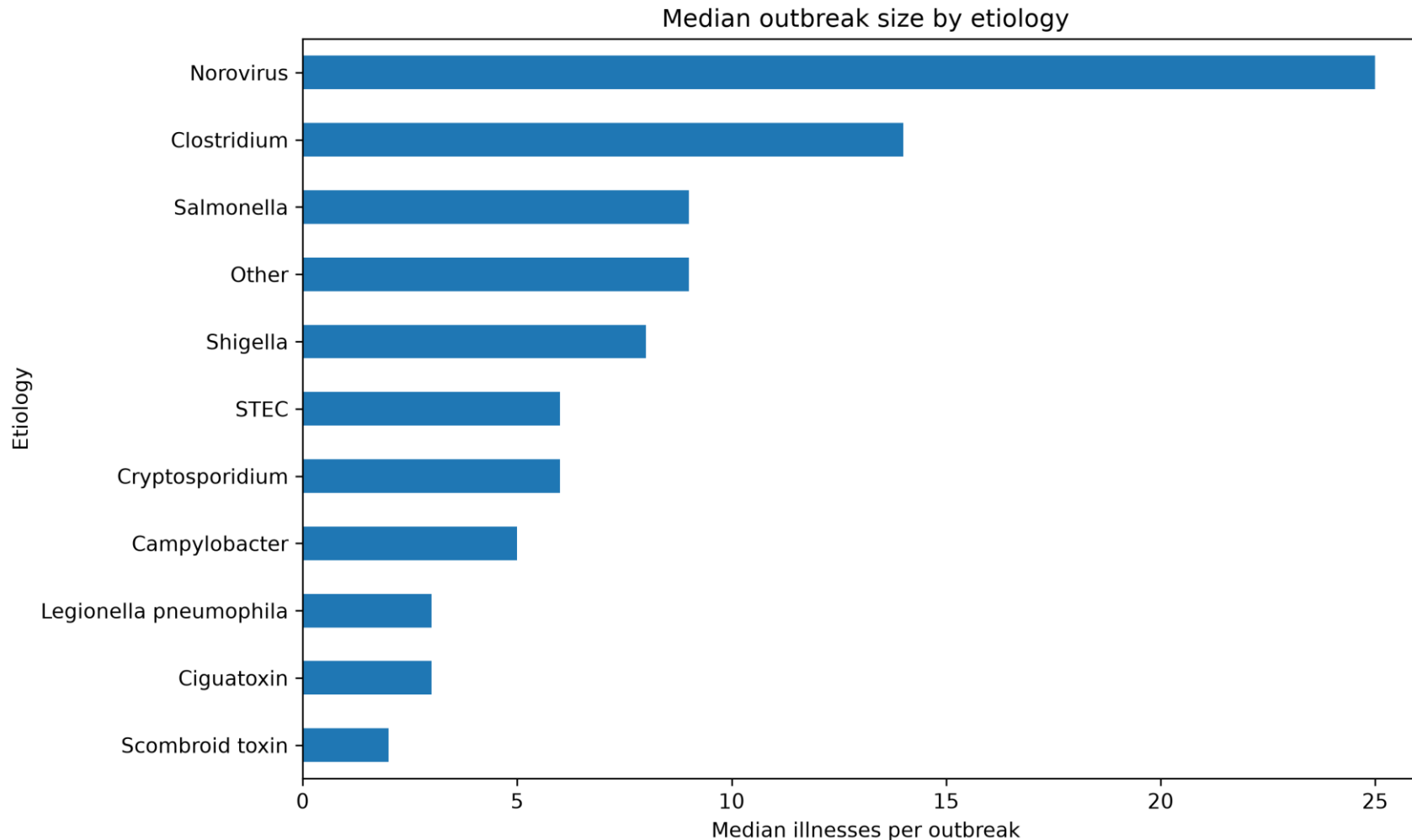




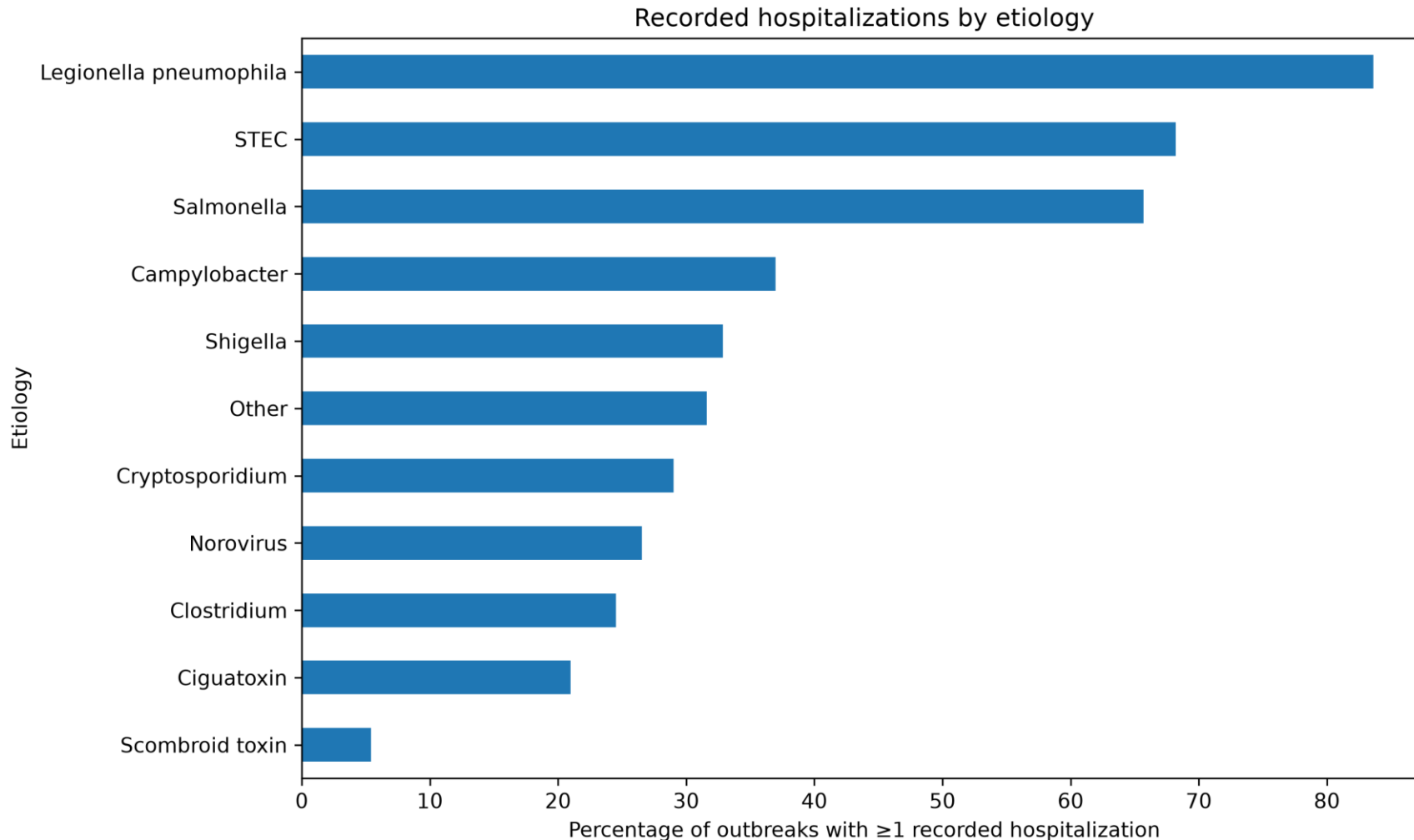
How did **outbreak settings** vary across the major outbreak etiologies?

- Outbreak settings varied considerably across etiologies
 - Norovirus → prominent in long-term care
 - Salmonella → prominent in restaurants and private residences
 - Some etiologies showed distinctive setting patterns
 - Other distinct patterns were also observed:
 - Approximately **54.4% of Shigella outbreaks** occurred in school settings
-

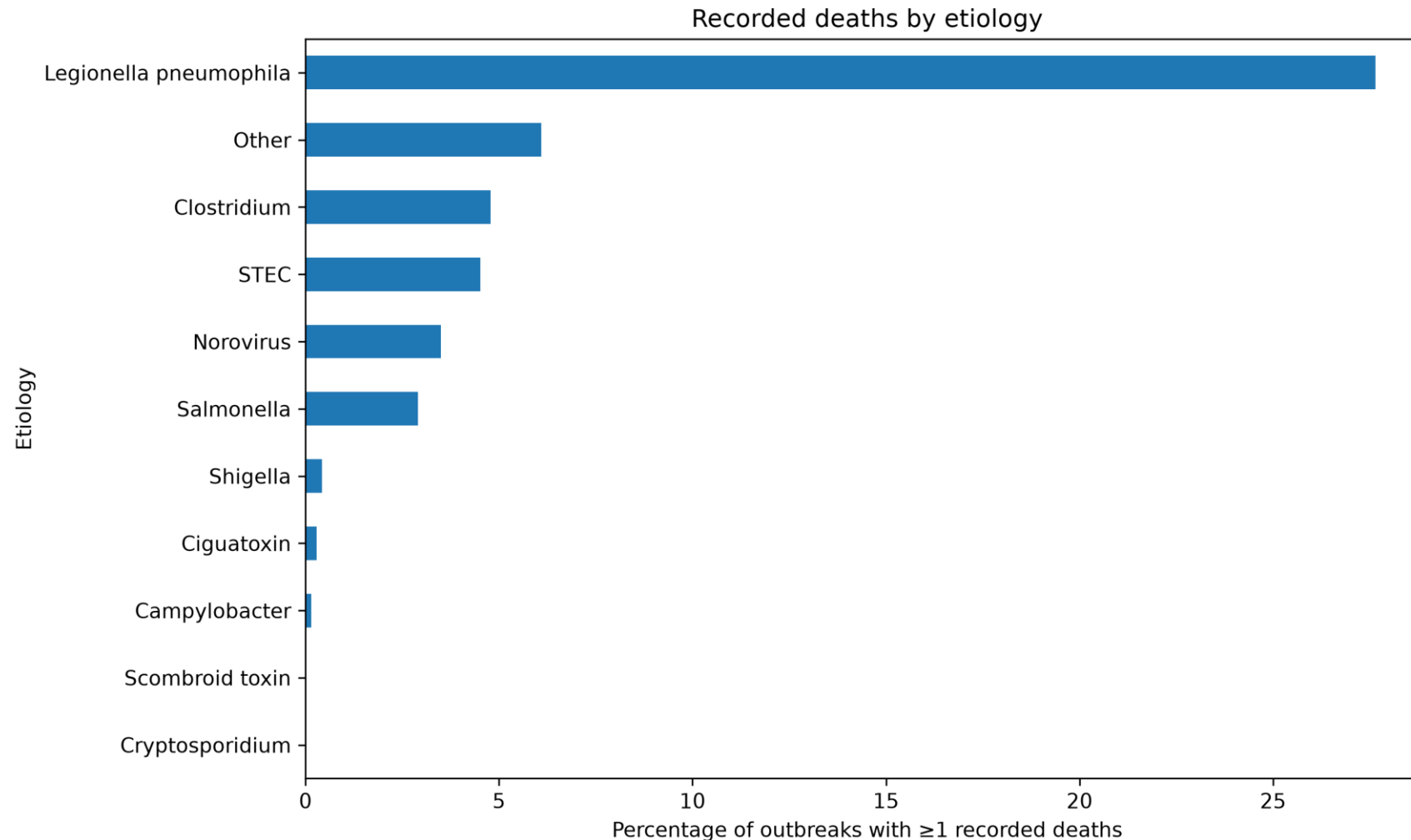
What was the typical outbreak size for each major etiology, based on the **number of reported illnesses per outbreak**?



How did the proportion of outbreaks with at least one recorded **hospitalization** vary across the major etiologies?



How did the proportion of outbreaks with at least one recorded **death** vary across the major etiologies?



Limitations

Reported outbreaks rather than all outbreaks

Restriction to single confirmed etiologies

Missing data

Etiology categorization

Descriptive analysis

Temporal interpretation



Recommendations

- Public-health surveillance and outbreak-response strategies should consider both the **frequency** and **characteristics of different etiologies**
 - **Norovirus** was the most frequently reported etiology (**52.8%**)
 - However, **Legionella pneumophila** showed much higher proportions of outbreaks with recorded **hospitalization (83.6%)** and **death (27.6%)**
 - Understanding which transmission routes, settings, and health outcomes are associated with specific etiologies can **help guide targeted prevention, infection-control, and food-safety measures**
-



Conclusion

- The analysis demonstrates that outbreak etiologies have distinct patterns across transmission, setting, outbreak size, hospitalization, and mortality
 - Examining these characteristics together provides a more informative picture of reported outbreaks than examining outbreak frequency alone
-

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THANK YOU

ANY QUESTIONS?

